

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Discuss the Role of chemical Engineer in chemical process industry. Write any five applications of chemical engineering. (CO1)

Q.24 Explain manufacturing process of urea with the help of neat and clean flowsheet (CO5)

Q.25 Write short notes on any four of the following.

- a) Crystallization (CO5)
- b) Drying (CO5)
- c) Reynold's number (CO1)
- d) Amagat's law (CO1)
- e) Heat Exchanger. (CO5)
- f) Rate of reaction

1st Sem / Chemical

Subject : Introduction to Chemical Engineering

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Formula for Density is (CO1)

- a) Mass/Volume b) Mass*Volume
- c) Volume/Mass d) None of these

Q.2 Mode of heat transfer in which energy is Transmitter by vibration of molecules. (CO2)

- a) Convection b) Conduction
- c) Radiation d) None of these

Q.3 In which of the following reaction, Heat energy is released during the process (CO4)

- a) Endothermic reaction
- b) Homogeneous Reaction
- c) Heterogeneous reaction
- d) Exothermic Reaction

Q.4 Formula for half-life of first order reaction is (CO4)

- a) $0.693 \times K$ b) $K/0.693$
c) $0.693/k$ d) K

Q.5 Mole fraction of constituents A in a mixture is defined as (CO1)

- a) Mole of A/Total moles in a mixture
b) Mole of A $\times$ total moles in a mixture
c) Weight of A/ Total weight of mixture
d) Weight of A/ molecular weight of A

Q.6 Equipment for drying operation is (CO5)

- a) Distillation column b) Tray Dryer
c) Crystallizer d) Cyclone Separator

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define black body. (CO2)

Q.8 List any two types of heat exchanger. (CO5)

Q.9 Define Chemical Engineering. (CO3)

Q.10 Define unit Operation. (CO1)

Q.11 Describe Homogeneous reactions. (CO4)

Q.12 List any two Types of Non-Newtonian Fluids. (CO1)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Define unit operation and unit process. (CO1)

Q.14 Explain conduction, convection and radiation with one example each. (CO2)

Q.15 Differentiate between reversible and irreversible reactions. (CO4)

Q.16 Describe Fick's law of diffusion and write the unit of diffusivity. (CO3)

Q.17 Differentiate between order of reaction and molecularity of reaction. (CO4)

Q.18 State Fourier's law of heat conduction and write the unit of thermal conductivity. (CO2)

Q.19 Define laminar flow and turbulent flow. (CO1)

Q.20 Explain viscosity and specific gravity, also mention the S.I. units. (CO1)

Q.21 State Dalton's law and Rault's law. (CO1)

Q.22 Differentiate between batch, semi-batch and continuous Process. (CO1)